EVALUATION SCHEME – MAT 2240 MID-TERM EXAMINATION

Multiple Choice Questions

1. What is the language accepted by the finite automaton given below? (1M)

Ans. (a) All non-empty strings beginning and ending in different symbols. — 1

2. Which of the following strings is not accepted by the finite automaton given below? (1M)

Ans. (a) 01001000 — 1

3. Which of the following languages is regular? (1M)

Ans. (a) All the words found in the Oxford English Dictionary. — 1

4. Which of the following regular expressions generates an infinite set? (1M)

Ans. (a) $a(a + bb + (a + b)^*)b$ — 1

5. A regular expression that generates all these strings (among others) is ab, aaaa ... (1M)

Ans. (a) $a(b + a(b + a(b + a(b + a)))$ — 1

6. For an interpretation i, we know that $i(p) = 1$ Then which of the following can be determined uniquely by i with this information only? (1M)

Ans. (a) $p \to (q \to p)$ — 1

7. Find the person who, in any case, is guilty. (1M)

Ans. (a) Jed — 1

Descriptive Questions

8. Determine whether the following consequences are valid: (3M)

   (a) $\neg(r \wedge \neg\neg q) \vdash (\neg q \vee \neg r)$

   (b) $p \vee \neg q, p \to \neg r \vdash q \to \neg r$

Ans. Consider the truth table below.

r	q	$\neg\neg q$	$r \wedge \neg\neg q$	$\neg(r \wedge \neg\neg q)$	$\neg q$	$\neg r$	$\neg q \vee \neg r$
1	1	1	1	0	0	0	0
1	0	0	0	1	1	0	1
0	1	1	0	1	0	1	1
0	0	0	0	1	1	1	1

Compare the columns:

- $\neg(r \wedge \neg\neg q)$
- $\neg q \vee \neg r$

They are identical in every row. Therefore, the conclusion follows from the premise. Hence the consequence is a **valid consequence**. — 1

p	q	r	$\neg q$	$p \vee \neg q$	$\neg r$	$p \to \neg r$	$q \to \neg r$
1	1	1	0	1	0	0	0
1	1	0	0	1	1	1	1
1	0	1	1	1	0	0	1
1	0	0	1	1	1	1	1
0	1	1	0	0	0	1	0
0	1	0	0	0	1	1	1
0	0	1	1	1	0	1	1
0	0	0	1	1	1	1	1

Rows where both premises $= 1$: Row 2, Row 4, Row 7, and Row 8

In all these rows $q \to \neg r = 1$.

Therefore, the consequence is **valid**. — 2

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9. State and prove the Reductio ad Absurdum theorem. (3M)

Ans. Statement: Let Σ be a set of propositions and let w be a proposition. Then

(i) $\Sigma \models w$ iff $\Sigma \cup \{\neg w\}$ is unsatisfiable.

(ii) $\Sigma \models \neg w$ iff $\Sigma \cup \{w\}$ is unsatisfiable.

Proof. (i) Let $\Sigma \models w$. Let i be any interpretation. If $i \models \Sigma$, then $i \models w$. So, i is not a model of $\neg w$.

Thus, i is not a model of $\Sigma \cup \{\neg w\}$. If i is not a model of Σ, then i is not a model of q for some $q \in \Sigma$; hence i is not a model of $\Sigma \cup \{\neg w\}$. In any case, i is not a model of $\Sigma \cup \{\neg w\}$. That is, $\Sigma \cup \{\neg w\}$ is unsatisfiable.

Conversely, let $\Sigma \cup \{\neg w\}$ be unsatisfiable. Let $i \models \Sigma$. Then i is not a model of $\neg w$; that is, $i \models w$.

Therefore, $\Sigma \models w$.

(ii) Take w instead of $\neg w$, and $\neg w$ instead of w in the proof of (i). □

Statement. — 1

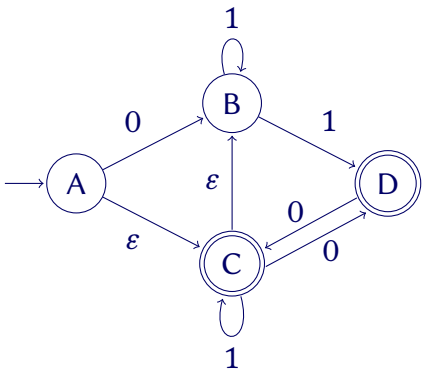
First part of the Proof for (i) — 1

Second part of the Proof for (i). — 1/2

Proof for (ii) — 1/2

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10. Convert the NFA given below to a regular expression. (5M)



Ans. GNFA 1 — 1

GNFA 2 — 1

GNFA 3 — 2

$(\varepsilon + 01^*10)(1(0 + 1^*1)0)^*(\varepsilon + 0 + 1^*1) + 01^*1$ — 1

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11. Show that the language $L = \{x_1x_2 \cdots x_{2n} | x_i \in \{0,1\}, x_{2n+1-i} = \overline{x_i}\}$ is not regular. (4M)

Ans. Let us assume for the sake of contradiction that L is regular. Let p be the pumping length given by the Pumping Lemma. — 1

Choose a string $s = 0^p1^p0^p1^p$. The string $s \in L$ because its second half (0^p1^p) is the reverse complement of its first half (0^p1^p). — 1

By the Pumping Lemma, s can be written as xyz where $|xy| \leq p$ and $|y| \geq 1$. Thus, y must consist entirely of 0s from the beginning of the string. — 1

If we pump y to xy^2z, the number of initial 0s increases, but the second half of the string remains unchanged. The resulting string no longer satisfies the condition that the second half is the reverse complement of the first. This is a contradiction, hence L is **not regular**. — 1

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12. Construct a pushdown automata that accepts the language $L = \{a^n b^{2n} | n \geq 0\}$. (4M)

Ans. Let $M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$ be a PDA where $\Gamma = \{A, Z_0\}$. — 1
The transitions δ are defined as follows:

1. $\delta(q_0, \epsilon, Z_0) = \{(q_f, Z_0)\}$ *(Accepts ϵ for $n = 0$)*
2. $\delta(q_0, a, Z_0) = \{(q_1, AAZ_0)\}$ *(Push two As for the first a)* — 1
3. $\delta(q_1, a, A) = \{(q_1, AAA)\}$ *(Push two As for every subsequent a)*
4. $\delta(q_1, b, A) = \{(q_2, \epsilon)\}$ *(Pop one A for the first b)* — 1
5. $\delta(q_2, b, A) = \{(q_2, \epsilon)\}$ *(Pop one A for all subsequent bs)*
6. $\delta(q_2, \epsilon, Z_0) = \{(q_f, Z_0)\}$ *(Accept if stack is empty of As at the end)* — 1

(i) Construct the NFA equivalent to the regular expression $(\mathrm{a} + \mathrm{bab})(\varepsilon + \mathrm{ba})^* + \mathrm{aba}$. (6M)

(ii) Using the subset construction, convert the NFA to a DFA.

(iii) Minimise the DFA.

Ans. (i) NFA — 1

(ii) DFA — 2

(iii) Minimal DFA with 6 states. — 3